Home SOC Lab – Splunk Installation & Deployment Handbook

This handbook documents the **minimal, clean, and correct procedure** to deploy **Splunk Enterprise on Ubuntu** and connect a **Windows 11 endpoint** using **Splunk Universal Forwarder** for a Home SOC lab.

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1. Lab Architecture (Simple & Practical)

Components

- **Splunk Enterprise (Ubuntu VM / Host)** – SIEM server
- **Windows 11 Machine** – Log source
- **Network** – Same LAN / NAT network

Ports Used

- `8000` – Splunk Web UI
- `8089` – Management (Forwarder ↔ Server)
- `9997` – Log receiving

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2. Splunk Enterprise Installation on Ubuntu

2.1 System Preparation

```
sudo apt update && sudo apt upgrade -y
sudo apt install curl wget -y
```

2.2 Download Splunk Enterprise

1. Create a free account on Splunk.
2. Download **Splunk Enterprise for Linux** (`.deb` or `.tgz`)
3. Or copy the direct download link:

```
wget -O splunk-10.0.2-e2d18b4767e9-linux-amd64.deb "https://download.splunk.com/products/splunk/releases/10.0.2/linux/splunk-10.0.2-e2d18b4767e9-linux-amd64.deb"
```

2.3 Install Splunk

If `.deb` file

```
sudo dpkg -i splunk-10.0.2-e2d18b4767e9-linux-amd64.deb
```

2.4 Start Splunk & Accept License

```
sudo /opt/splunk/bin/splunk start --accept-license
```

- Accept license
- Create **admin username & password**

2.5 Enable Auto Start (Important for Lab Stability)

```
sudo /opt/splunk/bin/splunkenable boot-start
```

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3. Accessing Splunk Web

Open browser:

- Local:

  ```
  http://127.0.0.1:8000
  ```

- Remote:

  ```
  http://<ubuntu_ip>:8000
  ```

Login with the admin credentials you created.

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4. Enable Log Receiving on Splunk

4.1 Configure Receiving Port

1. Go to **Settings → Forwarding and Receiving**
2. Under **Receive Data** → **Add New**
3. Set port:

   ```
   9997
   ```

4. Save

4.2 Create an Index

1. Go to **Settings → Indexes**
2. Create index:

   ```
   windows11
   ```

3. Save

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5. Splunk Universal Forwarder Installation (Windows 11)

5.1 Download

- Download **Splunk Universal Forwarder for Windows (.msi)**

5.2 Install Universal Forwarder

1. Run `.msi` installer
2. Accept license
3. Enter Splunk **admin credentials**
4. Configure Splunk Server:
   - **Server IP**: Ubuntu IP
   - **Management Port**: `8089`
   - **Receiving Port**: `9997`

5.3 Select Logs

Choose:

- Windows Event Logs (System, Security, Application)

This is sufficient for a **Home SOC lab**.

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6. Verify Forwarder Connection

On Splunk Server:

- Go to **Settings → Forwarder Management**
- You should see:
  - Forwarder connected
  - Hostname of Windows system

This confirms **successful deployment**.

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7. Verify Log Ingestion

Go to **Apps → Search & Reporting**

Run:

```
index=*
```

Or:

```
index=windows11
```

If logs appear → **Lab is operational**.

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8. What This Home SOC Lab Gives You

You now have:

- Centralized log collection
- Windows telemetry ingestion
- A base SIEM to practice:
  - Incident detection
  - Log analysis
  - Use-case creation
  - SOC workflows